MADBOX

HOME ASSIGNMENT

Welcome to Madbox's **Game Developer** Test 🤘🏻

Evaluation

This test is here to help us evaluate your hard skills in Game Development and in particular the following pillars: code **architecture**, code **readability**, code **performance**, and **game feel**.
When you are done we will review your project and will schedule a review session with you to discuss about it.
Do not exceed 15h of work on this project: keep things small, simple, clean. Everything you add into this game should demonstrate a different skill than what you've already put inside the project.

Goal

🎯 *You are to develop a game for a 3d character (the **hero**) in an RPG game with enemies.*

- Implement character movement in the 3d environment using 1 finger / Mouse left click + drag (reference: Archero)
- Enemies spawn in random positions on the map and move towards the hero to attack
  - There are 3 pre-made prefabs, you can modify them to fit your game. Each must have different stats.
- Hero will auto-attack when not moving.
  - Auto-select the closest target in range and sticks to it until the target is dead or the hero moves.
- Hero can equip 3 different weapons: curved sword, greatsword or longsword
  - When equipped, each weapon will modify:
    - Attack animation speed
    - Timing to apply damage to fit the animation
    - Hero's movement speed
    - Attack range
- It's up to you to decide how the hero selects a weapon.
- You **must use** the Unity Adressables package
- A **Game Loop** is required to play the game.

Live Ops

- We also shared a google Drive folder which contains:
  - 1 Google Sheet File with demo data
  - 1 url to fetch the GSheet as json data
- You must implement a Live Ops system in the game
  - You are free to choose the live ops content
  - Editing the GSheet should modify the game on the next session
  - Explain your choices and behavior in the README.md

Bonus

If you've got some time left, you can add any cool feature that can showcase any additional skill you've got.
Here are some **examples** of features you can do:

- **In game UI**
  - To display Health bars, scores... etc
- Enemies have a chance **drop loot** on death
  - It could be gold, xp, weapons...
- **Visual effects** on hit, death... etc
- **Editor tools**
- **Levels**
- **Cheats**

Materials

We included in the project:

- Assets for the hero
- Assets for the enemies
- Assets for the weapons
- A basic scene setup (MainScene)
- Addressables package is included
- TextMeshPro Essentials
- Unity Project is set up in a basic way using URP with Unity 2022.3.50f1

Delivery

When you are done, please send us your source code (via GitHub or any other online repository) so that we can make a code review.

The repository should also include a **README.md document** in which you'll mention

- How you approached this test: what were your different game making phases?
- The time you spent on each phase of the exercise
- The features that were difficult for you and why
- The features you think you could do better and how
- What you would do if you could go a step further on this game
- Live Ops system explanations
- Any comment you may have